claim	our assessment (an argument)
D answering	true in comparative form; no new inference principle; survives the objection from statistical practice
S logical form	an observation: textbooks are declarative; why that is good: open, relative to a reasoner
E explanation	true in the unification sense, with at least five gaps
L the ladder	SAI is on step 2, with partial results on step 3; step 4 has no formal handle
P progress	replaced by a list of formal differences; in the idealised setting only partiality concerns shape

Table 1: The argument map. The theorems that the assessments rest on are named in Section 1.1.

1 What a theory does and what it is

What does a scientific theory do? It may predict observations (Solomonoff, 1964); answer questions posed in some language (van Fraassen, 1980, ch. 5); explain (Hempel and Oppenheim, 1948); or solve problems (Laudan, 1977). What is it? A program that predicts observations; a set of sentences closed under consequence (Suppe, 1977); a class of models (Suppes, 1960; van Fraassen, 1980); a core of models with intended applications (Sneed, 1971; Balzer et al., 1987); or, on another line of thought, scientific knowledge rests on exemplars and tacit skills (Kuhn, 1970, 1977; Polanyi, 1958). The *axiom view* answers: questions posed in a given language, and a logical theory; the *raw-data view*: predictions of observations, and a program.

1.1 An argument map

We separate five claims (Table 1). **D**: a theory answers questions posed in a given language rather than predicting raw data (read comparatively). **S**: a theory is a logical theory, not a program. **E**: theories explain propositions. **L**: the ladder of **??**. **P**: SAI is progress over raw-data prediction, in what a theory does and in what it is.

What rests on theorems. Proved, in the stated settings: on data labelled true or false, the idealised formalism cannot see the difference between a theory and a program (Section 1.4); answering questions differs from raw-data prediction in target, data type, coherence and partiality, not in the inference principle (Propositions A.2, A.3, A.5 and A.6 and **????**; the theorems proved, this reading of them an argument); in the model of **??**, with a cost on the written theory, usefulness depends exponentially on the pair (reasoner, vocabulary), and on declarative against procedural form only by a constant factor once intermediate concepts are available and the reasoner can chain through them (**????**; open beyond that model); theoretical vocabulary is dispensable in the ideal mixture, while relations or entities pay for finite theories and under charged deduction (**????**); and "Kepler's laws are only approximately right" is a limitation of proof-requiring compatibility with deterministic theories, not of the axiom view as SAI formalises it: its mixture contains randomised theories, and in it an approximately true law is a law with a calibrated error model, carried by the randomness of the theory machine rather than stated as an axiom (**??**). Computed, conditional on the pool of forms and the codes: on noise-free interval data simplicity among compatible forms is language-relative, and with graded fit the data decide (**??**). A theorem plus an argued identification: Hempel's inductive-systematisation reply fails in the idealised setting (theoretical vocabulary changes the cumulative semimeasure loss on observational data by at most a constant number of bits, which depends on the universal machine and the setting; **??**). Arguments only: the descriptive observation; why declarative shape is good; that D survives the objection from statistical practice; claims about explanation beyond the toys (Proposition A.7 and **????**); that

the axiom view misses procedural, statistical and dynamic sides of theories; and our refusal, given a possibility result (**??**), to be realist about induced theories.

The ladder. We accept L. Step 2 is a step in the right direction, not the end point: an account of what theories do needs steps 3 and 4 as well. Step 4 is needed because theory-ladenness (Hanson, 1958) biases comparisons between theories that disagree about the base language: a theory that revises it is scored in a language it rejects.

1.2 The descriptive observation and the explanatory question

Claim S combines a description and an explanation. *The observation:* textbook presentations of fundamental theories are declarative, stating laws, definitions and models given by equations, not procedures that print predictions. Two qualifications. *Declarative is not sentential:* physics texts write something closer to a class of models specified by equations (Suppes, 1960, 1962) than to first-order axioms. The semantic view takes a theory to be such a class of models (on the relation of the two views, cf. Halvorson, 2012; Lutz, 2017); **??** shows where the choice of sentences enters our computation, and how much of its verdict survives without it. We use sentences because data are sentences and the prior of SAI is over texts; the question-answering view itself is neutral, since in partition semantics a question is a partition of logical space (cf. Groenendijk and Stokhof, 1984). *Theories in use have procedural parts,* such as numerical schemes and simulation codes, so the observation concerns textbook presentations, not every use of a theory.

The question, why the declarative shape is good and whether an ideal inducer should produce it, is open. Craig's construction rules out an answer from representability alone (Section 1.4); any answer must be relative to a reasoner with costs, or to a community of reasoners.

Question 1.1 (Why declarative shape?)**.** Is there a formal setting for a community of bounded reasoners who communicate, check and extend theories piece by piece, in which theories of declarative shape provably do better than programs?

A natural argument for S is that a theory answers in many directions while a program computes one cheaply; the second half holds of particular programs only: a program with internal variables answers every direction of a parity law on m variables with $O(m)$ operations. In the model of **??** directions matter only relative to a fixed weak reasoner and a cost on what is written or on the time to write it: for one parity law, forward chaining needs exactly m times as many rules as unit propagation needs clauses, over the base variables or with one round, and at most 9 times the written length of the least clausal theory with intermediate concepts and unbounded rounds (**??**).

1.3 What a theory does: questions against raw data

Questions are defined in **??**, as in the logic of questions (cf. Belnap and Steel, 1976; Hintikka, 1999); a theory *answers* a question $\{\varphi_1, \ldots, \varphi_k\}$ if it proves, with B, some φ_i, and *partially* if it proves a disjunction of some but not all of them (Definition A.1). A raw-data predictor is scored on one question, "what is the next observation?", and assigns probabilities to every event about future observations; it has no native notion of a question left open, and it answers a question about a law only as a question about all future data (item 5 of Table 2). SAI answers every question the base language can pose, and reports how much of its posterior answers none: the answer masses of a k-way question and its unanswered mass sum to 1 (Proposition A.2).

The conservation-law world. In a toy world a micro-state $x \in \{0,1\}^m$ moves by a uniformly random permutation of each weight class, so its weight is conserved; the raw data are the trajectory, the questions "what is $|x_t|$?". Until the trajectory returns to its start, the law saves a Bayes predictor of the raw data between $\frac{1}{2}\log\frac{\pi m}{2} + o(1)$ bits and one bit more per step (for $t \le Z/2$, Z the size of the weight class), a share $O(\log m/m)$ of its step loss; on $t_{\max}+1$ questions it saves, by construction, the share $t_{\max}/(t_{\max}+1)$ of the no-law loss. Yet by the chain rule the raw saving per step is the loss of the raw no-law predictor's own answer to "what is $|x_{t+1}|$?", and in each task a Bayes mixture pays at most one bit to learn the law (Proposition A.3). The tasks differ in their *target*, which losses are counted, not in what is learned or what the law is worth per step.

Answers never cost more description than raw modelling plus an *extractor* that reads them off the raw data: for a computable law μ of the raw data and an extractor η whose answers are μ-almost surely consistent with B, with μ_η the induced law of the answers, $L_A(D) \le -\log \mu_\eta(D) + K(\mu) + K(\eta) + c$, with c depending also on the family of question sentences (Proposition A.5). The extractor is the bridge from raw data to the question language, and its cost $K(\eta)$ is not small in real cases: it contains whatever recognises a planet in an image.

The objection from statistical practice. Much of physics confronts theory with near-raw data through generative models and information-theoretic losses, as in gravitational-wave searches, microwave-background cosmology and detector simulation. D survives in comparative form, we think, because these pipelines exist to answer a few questions (a mass, a spin, whether a particle exists); but the generative model is theory plus instrument plus noise, and the proof-requiring core of SAI, with no native noise model, sits far from this practice (**??**).

Purpose and logarithmic loss. The purpose fixes what is graded and with which tolerance, in SAI which propositions are data; scoring credences on them is a different job. The logarithmic score is, up to positive affine transformation, the only strictly proper local scoring rule on more than two outcomes (a result attributed to Bernardo, 1979), but which proper score is privileged is contested (Leitgeb and Pettigrew, 2010; Pettigrew, 2016), and we do not claim that it is; inside SAI even the loss convention can change the price of a datum without bound (**??**).

1.4 What a theory is, after Craig

The idealised formalism cannot see shape. Coin-free and one-tape, SAI $\asymp$ consistent SOI, by Craig's construction inside the universal prior (**????**). Adding vocabulary, restricting coin-free r.e. axiom induction to finite theories in a three-sorted extension, naming a reasoner, and reformulating the flagpole theories of Section 1.5, with any number of laws, each cost $O(1)$ bits (**????????** and Proposition A.9); the diagonal separations from SOI under time bounds come from free deduction, not from axioms (**??**). So, on data labelled true or false, the idealised formalism cannot see the difference between a theory and a program. Nor does D support S as a claim about what an ideal inducer should produce: labellers carry no theory and are equivalent to theories.

The one respect in which the idealised formalism separates theories from labellers is partiality. When "independent" is itself observed, a theory can leave a sentence open without committing to anything, while a labeller must output "independent" after finitely many steps, and the gap is unbounded on one datum (**??**); similarly, in the semimeasure convention a theory is certain of the consequences of the data and a consistent labeller never is (**??**). The witnesses of the first gap are arithmetical sentences independent of a background containing Q, and no r.e. family of

	difference	support
Advantages for claims D and E		
1	*Data type:* labelled sentences of a chosen language; questions are sets of sentences.	**??** and Def. A.1
2	*Logical coherence:* consequences of the data get posterior probability 1, unlike for consistent SOI in the semimeasure convention; Bel is monotone under entailment and closed under conjunction up to additive error.	**??** and Prop. A.6
3	*Partiality:* "independent" is a native third value (its mass is the semimeasure defect); a k-way question's answer masses and unanswered mass sum to 1.	**??** and Prop. A.2
4	*An observed "independent" label* separates theories from labellers by an unbounded amount on one datum (coin-free and one-tape; $\mathcal{L} \supseteq \mathcal{L}_{\mathrm{ar}}$, $\mathsf{Q} \subseteq \mathrm{Cn}(B)$).	**??**
5	*Laws have belief values:* $\mathrm{Bel}_D(\forall x\, \varphi)$ is not the probability that all future data conform.	**??**
6	*Answers cost at most raw modelling plus an extractor,* and can cost far less.	Props. A.3 and A.5
Neutral differences		
7	*Piecemeal confirmation:* credence in single laws; SAI inherits the Bayesian treatment of Duhem's problem (Dorling, 1979), which Strevens (2001) calls question-begging, and allocates blame by the prior, so by the syntax.	definitions; an argument
8	*Truth from unprovability:* if $\mathsf{Q} \subseteq \mathrm{Cn}(B)$, belief or independent mass on a Π_1 sentence certifies its truth; a curiosity of arithmetic data.	**??**
9	*Explanatory direction needs questions the language can pose:* interventional data separate causal theories that the old vocabulary cannot.	Prop. A.7(c), a toy
Liabilities		
10	*Random theories, false theories:* on the lottery stream bounded one-tape loss is achieved by random theories and by no consistent r.e. theory; in true arithmetic with true (adversarial) data the limit posterior can live on false theories, a possibility result.	**????**
11	*Language-relative simplicity, within bounds:* faithful translations change weights by $2^{\pm(K(\tau)+c)}$; within that bound the primitives in which laws are written decide real choices, such as the preferred orbit of Mars on noise-free interval data.	**??**; **??**

Table 2: What formally separates SAI from raw-data prediction. Items 7 and 9 and the sorting are arguments, the others results. Items 2 and 4 compare SAI with operator induction over consistent labellers, the program-shaped counterpart of a theory; they bear on claim S as well as on D.

them shows a large gap (**??**). We read this as a formal trace of partiality, one of the candidate advantages of logical form, not as support for claim S about empirical theories.

In the idealised setting, on data labelled true or false, the shape of a theory is invisible: every consistent labeller is an axiom system at a cost of $O(1)$ bits. In the settings we study, shape becomes visible only when something other than program length is charged (the written theory, the time to produce it, or the derivations) or the class of hypotheses is restricted (to finite theories); there it is relative to a reasoner and a vocabulary, and for theoretical terms what pays is relations and entities, not vocabulary as such (**??**). Whether this holds beyond these settings is open (**??**). We find it plausible that the declarative shape of written theories is good, and not merely common, but our results do not show it. Where shape is visible in our models, the large effects lie in the pair (reasoner, vocabulary), and declarative against procedural form costs at most a constant factor once intermediate concepts are available and the reasoner can chain through them (**??**); nothing in our results favours sentences over classes of models. What is missing is a formal setting in which that shape provably pays; communication and piecemeal checking by communities of reasoners are the candidates we see (Question 1.1).

1.5 Explanation

In the axiom view T explains φ when $B \cup T \vdash \varphi$, the deductive-nomological pattern (Hempel and Oppenheim, 1948), and an explanation is better the more a short theory derives, as in unification accounts (Friedman, 1974; Kitcher, 1981); the explanatory power of **??** gives one precise measure of this. But derivability is symmetric where explanation is not: a pole's height and the sun's elevation entail the shadow's length, and conversely (Bromberger, 1966, as attributed by Glymour, 1978). Formally (Proposition A.7): over real closed fields, four formulations of the shadow law, with the height or the reference shadow as the hub, are deductively equivalent, so the coin-free posterior keeps their prior odds on every compatible data set; under derivation charges which quantity is the hub is a property of the formulation, not of the world; and one interventional datum separates two causal versions that agree in the old vocabulary.

What the toys show. The explanation relation of SAI is direction-free. Kitcher's answer to the asymmetry (Kitcher, 1989), in its usual summary, appeals to other laws in which the height figures: with k of them, the formulation with the shadow as hub is longer by an amount growing with k in a literal code without definitions (if the reference shadow is not primitive or some objects cast no shadow), by a constant with one definition, and by $O(1)$ under the universal prior (Proposition A.9): Kitcher's answer is a whole-theory simplicity effect that SAI reproduces only under a listing code. Direction enters cleanly with interventional questions, but their semantics was written in by hand, so the toy is consistent with the interventionist account (Woodward, 2003) and gives it no support over rivals such as the causal-process account (Salmon, 1984).

We conclude that derivation from a short theory is explanation in the unification sense, with at least five gaps; we do not claim that derivation is explanation outright. The gaps are *asymmetry*, as above; *explanation by approximate correction*: Newton's theory does not entail Kepler's laws but, in an exact toy, proves the honest old law on every window and refutes the exact one (**??**), so explaining must include deriving an honest approximation with an error bound on a domain; *irrelevance* (hexed salt; a man taking birth-control pills): the length prior penalises an irrelevant axiom in a whole theory, but a derivation does not mark which axioms it used; the *conjunction problem* (L is derived from $L \wedge L'$), which the length prior and the explanatory power address in part; and *statistical explanation*: the approximate laws that SAI prefers come with error models (**??**) and make their data probable rather than derive them (cf. the statistical-relevance model of Salmon, 1971). Unification can register an explanatory gain without a gain in accuracy, as with Copernicus against Ptolemy (**??**).

1.6 What separates SAI from raw-data prediction

We replace claim P by the formal differences of Table 2. Neither the inference principle nor, in the idealised setting, the shape of hypotheses separates the two, except through partiality. This is the progress claim we believe: for what a theory does, SAI is a step in the right direction; for what a theory is, the idealised formalism is neutral on data labelled true or false.

SAI is Bayesian confirmation with a universal prior in which a theory either proves the data or is eliminated, and it inherits that framework's problems in specific forms: the dependence of the prior on the machine and the language, bounded by constants that are not small at the scale of our examples (**??????**); logical omniscience and the problem of old evidence (**??**); and the allocation of credit among the axioms of a theory (item 7 of Table 2).

A Proofs for what a theory does and is

This appendix proves the propositions of Section 1, and states in full the five that the section summarises in prose (Propositions A.3, A.5 to A.7 and A.9). Appendices A.1 to A.4 concern questions against raw data; Appendices A.5 and A.6 give the precise setting of the flagpole toys, prove Proposition A.7, and extend it to a system of laws.

A.1 The answer distribution

Questions, and the answers a theory proves, are defined in **??**; $[k] := \{1, \dots, k\}$.

Definition A.1 (Answers)**.** A theory T *answers* a question $\mathcal{Q} = \{\varphi_1, \dots, \varphi_k\}$ if $B \cup T \vdash \varphi_i$ for some i, and *answers it partially* if $B \cup T \vdash \bigvee_{i\in J} \varphi_i$ for some J with $\emptyset \neq J \subsetneq [k]$.

Proposition A.2 (The answer distribution)**.** *Let $M_A(D) > 0$, let $\mathcal{Q} = \{\varphi_1, \dots, \varphi_k\}$ be a question, and let $\mathrm{Un}_D(\mathcal{Q}) := \mathbb{U}(E_D \setminus \bigcup_i O_{\varphi_i})/\mathbb{U}(E_D)$, the posterior probability that the random theory proves no direct answer. (a) $\sum_{i\le k} \mathrm{Bel}_D(\varphi_i) + \mathrm{Un}_D(\mathcal{Q}) = 1$. (b) For $\emptyset \neq J \subsetneq [k]$, $\mathrm{Bel}_D(\bigvee_{i\in J} \varphi_i) \geq \sum_{i\in J} \mathrm{Bel}_D(\varphi_i)$, and the difference is the posterior probability that the random theory answers $\mathcal{Q}$ partially through J but proves no φ_i with $i \in J$; for $J = [k]$, $\mathrm{Bel}_D(\bigvee_i \varphi_i) = 1$ and the difference is $\mathrm{Un}_D(\mathcal{Q})$. (c) For $\mathcal{Q} = \{\varphi, \neg\varphi\}$, (a) is the trichotomy $\mathrm{Bel}_D(\varphi) + \mathrm{Dis}_D(\varphi) + \mathrm{Indep}_D(\varphi) = 1$ of **??**.*

Proof. On E_D the theory $B \cup A_\omega$ is consistent. If it proved φ_i and φ_j with $i \neq j$, it would prove $\varphi_i \wedge \varphi_j$, while $B \vdash \neg(\varphi_i \wedge \varphi_j)$; so it proves at most one direct answer. Hence the sets $E_D \cap O_{\varphi_i}$, $i \leq k$, are pairwise disjoint, and together with $E_D \setminus \bigcup_i O_{\varphi_i}$ they partition E_D. Dividing the measures by $\mathbb{U}(E_D) > 0$ gives (a), since $\mathrm{Bel}_D(\varphi_i) = \mathbb{U}(E_D \cap O_{\varphi_i})/\mathbb{U}(E_D)$.

(b) For $i \in J$, $O_{\varphi_i} \subseteq O_{\bigvee_{j\in J}\varphi_j}$, and the sets $E_D \cap O_{\varphi_i}$ are pairwise disjoint; so

$$\mathbb{U}\Big(E_D \cap O_{\bigvee_{j\in J}\varphi_j}\Big) = \sum_{i\in J} \mathbb{U}(E_D \cap O_{\varphi_i}) + \mathbb{U}\Big(E_D \cap O_{\bigvee_{j\in J}\varphi_j} \setminus \bigcup_{i\in J} O_{\varphi_i}\Big).$$

Divide by $\mathbb{U}(E_D)$: the last term is the posterior probability that the random theory proves $\bigvee_{j\in J}\varphi_j$, that is, answers $\mathcal{Q}$ partially through J, but proves no φ_i with $i \in J$. For $J = [k]$, $B \vdash \bigvee_i \varphi_i$, so $E_D \subseteq O_{\bigvee_i \varphi_i}$ and $\mathrm{Bel}_D(\bigvee_i \varphi_i) = 1$; by (a) the difference is $1 - \sum_i \mathrm{Bel}_D(\varphi_i) = \mathrm{Un}_D(\mathcal{Q})$.

(c) For $\mathcal{Q} = \{\varphi, \neg\varphi\}$ ($k = 2$), $\mathrm{Bel}_D(\neg\varphi) = \mathrm{Dis}_D(\varphi)$ and $\mathrm{Un}_D(\mathcal{Q})$ is the posterior probability that the random theory decides neither, which is $\mathrm{Indep}_D(\varphi)$. □

A.2 The conservation-law world

For even m, micro-states are $x \in \{0,1\}^m$ with weight $|x|$; the dynamics σ is a uniformly random permutation of each weight class, so the weight is conserved (the law); x_0 is uniform on the class of weight $m/2$, of size $Z := \binom{m}{m/2}$, and $x_{t+1} := \sigma(x_t)$. The raw predictors Raw^+, Raw^- are Bayes for "σ is a uniformly random permutation of each weight class", resp. "of $\{0,1\}^m$", and their step-t loss is that of predicting x_{t+1} from $x_0, \dots, x_t$. The questions are "what is $|x_t|$?", $t = 0, 1, \dots$; the question predictors Qa^+, Qa^-, $\mathsf{Qa}^{\mathrm{bin}}$ are Bayes for "$|x_t| = |x_0|$, with $|x_0|$ uniform on $\{0, \dots, m\}$", "the $|x_t|$ are i.i.d. uniform on $\{0, \dots, m\}$" and "the $|x_t|$ are i.i.d. binomial$(m, \frac{1}{2})$". In each task Mix gives prior $\frac{1}{2}$ to each of the task's two hypotheses ($\mathsf{Raw}^\pm$, resp. $\mathsf{Qa}^\pm$). Let t^* be the length of the cycle of σ through x_0.

Proposition A.3 (The conservation-law world)**.** *(a) For $t < t^*$ the step-t losses of Raw^+ and Raw^- are $\log(Z-t)$ and $\log(2^m-t)$; for $t \geq t^*$ both are 0. The law's gain $\kappa_t := \log \frac{2^m-t}{Z-t}$ $(t < t^*)$ is nondecreasing in t; for $t \leq Z/2$, $\log \frac{2^m}{Z} \leq \kappa_t \leq \log \frac{2^m}{Z} + 1$, where $\log \frac{2^m}{Z} = \frac{1}{2} \log \frac{\pi m}{2} + o(1)$, and the share $\kappa_t / \log(2^m - t)$ of the step loss is $O(\log m/m)$. (b) Qa^+ pays $\log(m+1)$ for $|x_0|$ and 0 afterwards, Qa^- pays $\log(m+1)$ per question; over $t_{\max}+1$ questions the law's share of the no-law loss is $t_{\max}/(t_{\max}+1)$. (c) In each task the cumulative loss of Mix exceeds that of the better of its two hypotheses by at most one bit, on every trajectory. (d) For $t < t^*$, $\log(2^m - t) = \kappa_t + \log(Z-t)$ with $\kappa_t = -\log P_{\mathsf{Raw}^-}(|x_{t+1}| = m/2 \mid x_0, \ldots, x_t)$: the law's raw gain is exactly the loss of Raw^-'s own answer to "what is $|x_{t+1}|$?". The loss of $\mathsf{Qa}^{\mathrm{bin}}$ on each question is κ_0.*

Lemma A.4 (Next state of a random permutation)**.** *Let σ be a uniformly random permutation of a finite set X, and let $x_0, \ldots, x_t \in X$ be distinct with $\sigma(x_s) = x_{s+1}$ for $s < t$. Conditionally on these t values of σ, $\sigma(x_t)$ is uniform on $(X \setminus \{x_0, \ldots, x_t\}) \cup \{x_0\}$, a set of $|X| - t$ elements.*

Proof. Conditionally on $\sigma(x_s) = x_{s+1}$ for $s < t$, the restriction of σ to $X \setminus \{x_0, \ldots, x_{t-1}\}$ is a uniformly random bijection onto $X \setminus \{x_1, \ldots, x_t\}$, since every such bijection extends the given values to exactly one permutation. So $\sigma(x_t)$ is uniform on $X \setminus \{x_1, \ldots, x_t\} = (X \setminus \{x_0, \ldots, x_t\}) \cup \{x_0\}$, which has $|X| - t$ elements. □

Proof of Proposition A.3. (a) For $t < t^*$ the states $x_0, \ldots, x_t$ are distinct, all of weight $m/2$, and x_{t+1} is either unvisited or x_0. Under the hypothesis of Raw^+ the permutations of the weight classes are independent, and the trajectory stays in the class of x_0; Lemma A.4 with X that class ($|X| = Z$) gives the predictive probability $1/(Z-t)$ of the true x_{t+1}, a step loss of $\log(Z-t)$. Under the hypothesis of Raw^- the lemma with $X = \{0,1\}^m$ gives $\log(2^m - t)$. For $t \geq t^*$ the path has returned to x_0, the cycle through x_0 is known to both predictors, and later states are determined: both losses are 0.

Since $2^m > Z \geq t^* > t$, the map $t \mapsto (2^m - t)/(Z-t)$ is increasing on $[0, Z)$ (its derivative is $(2^m - Z)/(Z-t)^2 > 0$), so κ_t is nondecreasing and $\kappa_t \geq \kappa_0 = \log(2^m/Z)$. For $t \leq Z/2$, $(2^m - t)/(Z-t) \leq 2^m/(Z/2)$, so $\kappa_t \leq \log(2^m/Z) + 1$. By Stirling's formula, $Z = \binom{m}{m/2} = 2^m \sqrt{2/(\pi m)}\,(1 + o(1))$, so $\log(2^m/Z) = \frac{1}{2} \log \frac{\pi m}{2} + o(1)$. Finally $\kappa_t \leq \log(2^m/Z) + 1 = O(\log m)$, and $\log(2^m - t) \geq \log(2^m - Z/2) \geq m - 1$ since $Z \leq 2^m$; so the share is $O(\log m/m)$ for $t \leq Z/2$.

(b) Qa^+ predicts $|x_0|$ uniformly on $\{0, \ldots, m\}$, a loss of $\log(m+1)$, and then $|x_t| = |x_0|$ with probability 1, which is true since the weight is conserved: loss 0. Qa^- pays $\log(m+1)$ on each question. Over the questions $t = 0, \ldots, t_{\max}$ the losses are $\log(m+1)$ and $(t_{\max}+1)\log(m+1)$, and the law saves the share $t_{\max}/(t_{\max}+1)$ of the latter.

(c) In each task Mix is the Bayes mixture with prior $\frac{1}{2}$ on the task's two hypotheses; x_0 has the same law under both raw hypotheses, so the posterior after x_0 is still $(\frac{1}{2}, \frac{1}{2})$. By the chain rule, the cumulative loss of a Bayes mixture is $-\log$ of the mixture probability of the whole sequence. If the two hypotheses give the sequence probabilities 2^{-a} and 2^{-b}, the mixture's loss is $-\log(\frac{1}{2}2^{-a} + \frac{1}{2}2^{-b}) \leq -\log(\frac{1}{2}2^{-\min(a,b)}) = \min(a,b) + 1$, on every trajectory.

(d) Under Raw^-, by Lemma A.4, x_{t+1} is uniform on a set of $2^m - t$ states, of which exactly $Z - t$ have weight $m/2$: the $Z-(t+1)$ unvisited states of the class of x_0, and x_0. So $P_{\mathsf{Raw}^-}(|x_{t+1}| = m/2 \mid x_0, \ldots, x_t) = (Z-t)/(2^m - t) = 2^{-\kappa_t}$, and conditionally on that weight x_{t+1} is uniform on those $Z - t$ states, which is exactly the predictive distribution of Raw^+. The chain rule splits the step loss $\log(2^m - t)$ of Raw^- into κ_t, the loss of its prediction of the weight, and $\log(Z-t)$, the loss of Raw^+. Finally, every $|x_t|$ equals $m/2$, and the binomial probability of $m/2$ is $Z2^{-m}$, so $\mathsf{Qa}^{\mathrm{bin}}$ pays $\log(2^m/Z) = \kappa_0$ on each question. □

Computations. A computation (code in the accompanying repository, seeded) gives the following. At the start of a trajectory the law's gain per raw step is 1.871, 2.148, 2.348, 2.837, 3.331, 3.829, 5.326 and 10.292 bits at $m = 8$, 12, 16, 32, 64, 128, 1024 and 10^6, against $\frac{1}{2}\log\frac{\pi m}{2} = 1.826$, 2.118, 2.326, 2.826, 3.326, 3.826, 5.326, 10.292; its share of the raw step loss is 0.23, 0.18, 0.15, 0.089, 0.052, 0.030, 0.0052 and 1.0×10^{-5}. The value $\kappa_{Z/2}$ is 2.659, 2.976, 3.199, 3.732, 4.258, 4.777 and 6.308 bits at $m = 8, \ldots, 1024$; the loss of Qa^- per question is 3.170, 3.700, 4.088, 5.044, 6.022, 7.011 and 10.001 bits, about twice κ_0. At $m = 12$ ($Z = 924$), averaging exactly over the cycle length (uniform on $\{1, \ldots, Z\}$), the expected raw loss over 300 steps is 2415.6 bits with the law and 3004.9 without (share 0.196), and the mixture's is 2416.6; over the 301 questions the losses are 3.70 and 1113.8 bits (share 0.997). Over a whole cycle the raw share is 0.232 and the mean gain 2.75 bits per step. In 2000 simulated runs the mean raw losses are within one standard error of these values, and the mixture never exceeds the better hypothesis by more than one bit. The identity of (d) holds at $m = 12$ for every $t < Z$ to within 1.8×10^{-15} bits in floating point.

A.3 Questions from raw data and an extractor

Answers never cost more description than raw modelling plus an *extractor* that reads them off the raw data. Let $\mathbb{X}$ be a finite alphabet, μ a computable probability measure on $\mathbb{X}^{\mathbb{N}}$ (the raw world), $(\varphi_q)_{q\in\mathbb{N}}$ a fixed computable family of sentences (the questions), and η a partial computable function with $\eta(q, x) \in \{0, 1\}$ computed from a finite prefix of x; put $f_x(\varphi_q) := \eta(q, x)$ and $\mu_\eta(D) := \mu\{x : f_x \supseteq D\}$.

Proposition A.5 (Raw modelling plus an extractor)**.** *If f_x is consistent with B for μ-almost every x, then $M_A(D) \geq 2^{-K(\mu)-K(\eta)-c}\mu_\eta(D)$ for every finite data set D of question–answer pairs, so $L_A(D) \leq -\log \mu_\eta(D) + K(\mu) + K(\eta) + c$; here c depends also on the family (φ_q). For a computable sequence x with f_x consistent with B, $L_A(D) \leq K(x) + K(\eta) + c$ on every D with $f_x \supseteq D$.*

Proof. Sampling. There is a partial computable map Ψ from tapes to $\mathbb{X}^{\mathbb{N}}$, computable from an index of μ, that is defined for $\mathbb{U}$-almost every tape and satisfies $\Psi_*\mathbb{U} = \mu$. This is the standard construction: read tape bits to refine a dyadic interval, and output the next symbol s as soon as the interval lies inside the μ-interval of s given the symbols already output; it fails only on the null set of tapes that hit interval boundaries. Each output symbol depends on a finite prefix of the tape.

The theory machine. Let G be the theory machine that, on a tape ω, computes $\Psi(\omega)$ as far as needed and, for each q, emits $\varphi_q^{\eta(q,\Psi(\omega))}$ once the tape prefix read so far fixes a prefix of $\Psi(\omega)$ on which the computation of $\eta(q, \cdot)$ halts. Each emission depends on a finite prefix of ω, so G is a c.e. set of pairs (prefix, sentence) as **??** requires, and $K(G) \leq K(\mu) + K(\eta) + c$, where c includes an index of the fixed family (φ_q). If $\Psi(\omega)$ is defined, then $A_\omega^G = \Gamma_{f_{\Psi(\omega)}}$, the set of labelled sentences of the labelling $f_{\Psi(\omega)}$.

The bound. Let H be the set of x for which f_x is consistent with B; $\mu(H) = 1$ by hypothesis. On every tape ω with $\Psi(\omega)$ defined, $\Psi(\omega) \in H$ and $f_{\Psi(\omega)} \supseteq D$, the theory $B \cup A_\omega^G$ is consistent and proves ψ_D, so $\omega \in E_D^G$. Hence $\nu_G(D) \geq \mathbb{U}\{\omega : \Psi(\omega) \in H,\ f_{\Psi(\omega)} \supseteq D\} = \mu\{x \in H : f_x \supseteq D\} = \mu_\eta(D)$. By **??**, $M_A(D) \geq 2^{-K(\nu_G)}\nu_G(D) \geq 2^{-K(G)}\nu_G(D) \geq 2^{-K(\mu)-K(\eta)-c}\mu_\eta(D)$, and taking $-\log$ gives the bound on L_A. For a computable sequence x, take $\mu = \delta_x$, with $K(\delta_x) \leq K(x) + c$; the hypothesis is that f_x is consistent with B, and $\mu_\eta(D) = 1$ exactly when $f_x \supseteq D$. □

The constant depends on the question family because the theory machine must write the sentences φ_q. The converse of the proposition fails badly, since answers do not determine raw

data; in the conservation-law world the raw loss exceeds the question loss by the whole micro-entropy (Proposition A.3).

A.4 Answering and predicting

The following gives a formal vocabulary for "answering questions is more than predicting data": answering implies predicting, (a), but not conversely, (b), and success at predicting data that present a theory does not settle the questions the data leave open, (c). Part (d) is the coherence that a predictor of separate labels lacks.

Proposition A.6 (Answering and predicting)**.** *(a) If a consistent r.e. theory T proves every datum of a consistent reveal sequence, the cumulative loss of M_A is at most $-\log \mathbb{U}_A(\mathrm{Ext}T) \le K_{\equiv}(T) + c$ in the semimeasure convention, in any order. (b) On the lottery stream of* **??***, M_A has bounded loss in the semimeasure convention, while no consistent r.e. theory proves every datum and $W_A(D_n) \to 0$. (c) On two-valued data presenting an incomplete consistent r.e. theory T^*, the limit posterior is the prior restricted to the consistent extensions of T^*, and on every sentence independent of $B \cup T^*$ all three values* Bel, Dis, Indep *have positive limits; three-valued data identify the class of T^*. (d) For all D with $M_A(D) > 0$ and all sentences,* $\mathrm{Bel}_D(\bigwedge_{i\le k}\varphi_i) \ge 1 - \sum_{i\le k}(1 - \mathrm{Bel}_D(\varphi_i))$*, and* $\mathrm{Bel}_D(\psi) \ge \mathrm{Bel}_D(\varphi)$ *whenever $B \cup \{\varphi\} \vdash \psi$. So the sentences with belief at least $1 - \delta$ are closed under entailment and, with error $k\delta$, under k-fold conjunction.*

Proof of Proposition A.6. (a) By **??**(d) the cumulative loss in the semimeasure convention is $L_A(D_n) - L_A(\emptyset)$, in any order, and by **??** it is at most $-\log \mathbb{U}_A(\mathrm{Ext}T) \le K_{\equiv}(T) + c$.

(b) In the setting of **??**, part (a) of that theorem bounds $M_A(D_n)$ below by a positive constant independent of n, so the cumulative loss $L_A(D_n) - L_A(\emptyset)$ is bounded; part (b) says that no consistent r.e. theory proves every datum (with B); and part (c) gives $W_A(D_n) \to 0$.

(c) This is **??**(b), (c): if $B \cup T^*$ is incomplete and $\mathrm{Cn}(D^{\pm}_{\infty}) = \mathrm{Cn}(T^*)$, the limit posterior is the prior restricted to the consistent extensions of T^*, and for every ψ independent of $B \cup T^*$ the limits of $\mathrm{Bel}_{D_n}(\psi)$ and $\mathrm{Dis}_{D_n}(\psi)$ lie in $(0,1)$ while $\lim_n \mathrm{Indep}_{D_n}(\psi) \ge \mathbb{U}_A[T^*]/\mathbb{U}_A(\mathrm{Ext}T^*) > 0$; and if every sentence appears with its T^*-trichotomy label, the posterior probability of the class of T^* tends to 1.

(d) A theory that proves φ and ψ proves $\varphi \wedge \psi$, so $O_\varphi \cap O_\psi \subseteq O_{\varphi\wedge\psi}$. Within E_D,

$$\mathbb{U}(O_\varphi \cap O_\psi \cap E_D) \ge \mathbb{U}(O_\varphi \cap E_D) + \mathbb{U}(O_\psi \cap E_D) - \mathbb{U}(E_D);$$

dividing by $\mathbb{U}(E_D)$ gives $\mathrm{Bel}_D(\varphi \wedge \psi) \ge \mathrm{Bel}_D(\varphi) + \mathrm{Bel}_D(\psi) - 1$, and induction on k gives $\mathrm{Bel}_D(\bigwedge_{i\le k}\varphi_i) \ge 1 - \sum_{i\le k}(1 - \mathrm{Bel}_D(\varphi_i))$. If $B \cup \{\varphi\} \vdash \psi$, then $O_\varphi \subseteq O_\psi$, so $\mathrm{Bel}_D(\psi) \ge \mathrm{Bel}_D(\varphi)$. □

A.5 The flagpole

Setting. The language has two sorts, poles and reals; B is the theory of real closed fields on the real sort. There are pole constants $p_1, \dots, p_{N_{\mathrm{p}}}$, a function symbol ht from poles to reals (the height) and a function symbol sh from poles and reals to reals ($\mathrm{sh}(p,u)$ is the length of the shadow of p when the tangent of the sun's elevation is $u > 0$). Data are sentences $\mathrm{sh}(p_i, u) = v$ and $\mathrm{ht}(p_i) = v$ with rational numerals u, v; interval versions behave in the same way. The two formulations of the shadow law are

$$\Lambda_h := \forall p\, \forall u\, \big(u > 0 \to \mathrm{sh}(p,u) \cdot u = \mathrm{ht}(p)\big),$$
$$\Lambda_s := \forall p\, \forall u\, \big(u > 0 \to \mathrm{sh}(p,u) \cdot u = \mathrm{sh}(p,1)\big) \ \wedge\ \forall p\, \mathrm{ht}(p) = \mathrm{sh}(p,1),$$

and, with rational numerals a_i for the heights, the four theories are

$$T_H := \Lambda_h + \{\mathrm{ht}(p_i) = a_i\}_{i \le N_\mathrm{p}}, \qquad T_S := \Lambda_h + \{\mathrm{sh}(p_i, 1) = a_i\}_{i \le N_\mathrm{p}},$$
$$T'_H := \Lambda_s + \{\mathrm{ht}(p_i) = a_i\}_{i \le N_\mathrm{p}}, \qquad T'_S := \Lambda_s + \{\mathrm{sh}(p_i, 1) = a_i\}_{i \le N_\mathrm{p}}.$$

In T_S the shadow at 45° "explains" the height. The *intervention vocabulary* consists of $\mathrm{sh}_H(p, u, v)$, the shadow length after the height of p is set to v, and $\mathrm{ht}_S(p, v)$, the height after the reference shadow $\mathrm{sh}(p, 1)$ is set to v by an intervention on the shadow (for instance, by an obstacle). The causal extensions, with the height exogenous and with the shadow exogenous, are

$$C_H := T_H + \{\forall p\, \forall u\, \forall v\, (u > 0 \to \mathrm{sh}_H(p, u, v) \cdot u = v),\ \forall p\, \forall v\, \mathrm{ht}_S(p, v) = \mathrm{ht}(p)\},$$
$$C_S := T_S + \{\forall p\, \forall u\, \forall v\, \mathrm{sh}_H(p, u, v) = \mathrm{sh}(p, u),\ \forall p\, \forall v\, \mathrm{ht}_S(p, v) = v\}.$$

Derivation charges. The cost of a derivation is the number of closed instances it uses of the universally quantified conjuncts of the law axioms (Λ_h has one such conjunct, Λ_s two); steps in B and uses of the initial conditions are free. The *least cost* of a datum for a theory is the least number of such instances that, together with B and the theory's initial conditions, entail it. By completeness this is the least cost of a derivation, so the count is semantic, as in Herbrand's theorem.

Proposition A.7 (The flagpole)**.** *(a) T_H, T_S, T'_H, T'_S are deductively equivalent over B; so every data set is compatible with all four or with none, in the coin-free version the posterior odds of any two programs for them equal their prior odds on every data set compatible with them, and $|K(T_S) - K(T_H)| \le c$. (b) Under derivation charges, every shadow datum $\mathrm{sh}(p_i, u) = v$ with $u \neq 1$ that they entail needs exactly one instance from T_H and T'_S, whose initial conditions sit at the hub of their formulation, and exactly two from T_S and T'_H: which quantity is the hub is a property of the formulation, not of the world. (c) C_H and C_S have the same consequences in the old vocabulary, and one interventional datum separates them.*

The precise form of part (b) is as follows. For a shadow datum $\mathrm{sh}(p_i, u) = a_i/u$ with $u \neq 1$, the least costs for T_H, T_S, T'_H, T'_S are 1, 2, 2, 1; for a height datum $\mathrm{ht}(p_i) = a_i$ they are 0, 1, 0, 1; and for a reference-shadow datum $\mathrm{sh}(p_i, 1) = a_i$ they are 1, 0, 1, 0. So the two hub theories T_H and T'_S have equal costs on all shadow data with $u \neq 1$. Their law texts differ by the constant $|\Lambda_s| - |\Lambda_h|$, and in a literal code (Appendix A.6) their initial conditions differ by $N_\mathrm{p}\Delta_\mathrm{sh}$, where $\Delta_\mathrm{sh} := |\mathrm{sh}(x, 1)| - |\mathrm{ht}(x)|$ is the extra length of the reference-shadow term; this is 0 if the language has a primitive for the reference shadow, and then the two texts differ by $|\Lambda_s| - |\Lambda_h|$, independently of N_p and the data. The precise form of (c) is: C_H and C_S have the same consequences in the language without sh_H and ht_S; $C_H \vdash \mathrm{sh}_H(p_1, 1, 2a_1) = 2a_1$ ("doubling the pole doubles its 45° shadow"), while $C_S \vdash \mathrm{sh}_H(p_1, 1, 2a_1) = a_1$, which refutes it when $a_1 \neq 0$; and $C_H \vdash \mathrm{ht}_S(p_1, v) = a_1$ while $C_S \vdash \mathrm{ht}_S(p_1, v) = v$, for every numeral v.

Lemma A.8 (Least costs in a star)**.** *Let T be one of the theories of this appendix without the dark poles: law axioms whose universally quantified conjuncts have closed instances of the form $\pi \to s_1 = \rho \cdot s_2$, with π a condition on real numerals, s_1, s_2 terms of one pole and ρ a real term that B proves nonzero when π holds, and one initial condition $s(p_i) = a_i$ per pole. For a pole p_i, let the* instance graph *have as vertices the closed terms $\mathrm{ht}(p_i)$, $\mathrm{sh}(p_i, u)$ (terms whose real arguments B proves equal are identified) and, in Appendix A.6, $\mathrm{qt}_j(p_i)$, and an edge for each closed instance whose condition B proves, joining the two terms it relates (an instance that relates a term to itself is trivial and is ignored). If the instance graph is a tree, the least cost of a datum $s'(p_i) = v$ that T entails is the distance in the graph from $s'(p_i)$ to the vertex fixed by the initial condition.*

Proof. The instances along the path from the fixed vertex to $s'(p_i)$, together with B and the initial condition, determine the value of $s'(p_i)$; since T entails the datum, that value is v, so the distance is an upper bound. Conversely, let I be a set of fewer instances than the distance. As the graph is a tree, the edges of I do not connect $s'(p_i)$ to the fixed vertex. Fix a model of T over $\mathbb{R}$ (each theory of this appendix has one). Build a model of B, the initial conditions and I over $\mathbb{R}$ as follows: on the connected component (for the edges of I) of the fixed vertex, and on every term of another pole, keep the values of the fixed model; on the component of $s'(p_i)$, give $s'(p_i)$ a value other than v and propagate it along the edges of I, which is consistent because the component is a tree and each edge is an invertible linear relation; give all other terms arbitrary values. Instances with a false condition hold trivially. This model falsifies the datum, so I does not entail it. □

Proof of Proposition A.7. (a) At $u = 1$, Λ_h gives $\mathrm{sh}(p, 1) = \mathrm{ht}(p)$; given this, $\mathrm{sh}(p, u) \cdot u = \mathrm{ht}(p)$ and $\mathrm{sh}(p, u) \cdot u = \mathrm{sh}(p, 1)$ are equivalent. So $B \vdash \Lambda_h \to \Lambda_s$; conversely Λ_s gives $\mathrm{ht}(p) = \mathrm{sh}(p, 1)$ and hence Λ_h. Given either law, the initial conditions $\mathrm{ht}(p_i) = a_i$ and $\mathrm{sh}(p_i, 1) = a_i$ are interderivable, so the four theories are deductively equivalent over B. Compatibility of a theory with a data set (consistent with B and proving every datum) depends only on its deductive closure, so every data set is compatible with all four or with none. In the coin-free version the posterior weight of a program is $2^{-|p|}$ times the indicator of compatibility, divided by the normaliser, so the odds of two compatible programs are their prior odds. The map on axiom lists that replaces each $\mathrm{ht}(p_i) = a_i$ by $\mathrm{sh}(p_i, 1) = a_i$, and its inverse, are fixed computable maps, so $|K(T_S) - K(T_H)| \le c$.

(b) For each pole, the instances of Λ_h join $\mathrm{ht}(p_i)$ to each $\mathrm{sh}(p_i, u)$ (a star centred at the height), and the instances of Λ_s join $\mathrm{sh}(p_i, 1)$ to each $\mathrm{sh}(p_i, u)$ with $u \neq 1$ and to $\mathrm{ht}(p_i)$ (a star centred at the reference shadow); instances with $u \le 0$ have false conditions, and instances at other poles are not connected to p_i. Both graphs are trees, and Lemma A.8 gives the counts: in the height-centred star with the height fixed (T_H), shadows and the reference shadow are at distance 1; with the reference shadow fixed (T_S), a shadow with $u \neq 1$ is at distance 2 and the height at 1; in the shadow-centred star with the height fixed (T'_H), a shadow with $u \neq 1$ is at distance 2 and the reference shadow at 1; with the reference shadow fixed (T'_S), shadows and the height are at distance 1. The law texts of T_H and T'_S are Λ_h and Λ_s; their initial conditions differ only in the terms $\mathrm{ht}(p_i)$ and $\mathrm{sh}(p_i, 1)$, once per pole.

(c) Without the intervention symbols, C_H and C_S are T_H and T_S plus axioms about fresh symbols. Every model of T_H expands to a model of C_H (put $\mathrm{sh}_H(p, u, v) := v/u$ for $u > 0$, arbitrary otherwise, and $\mathrm{ht}_S(p, v) := \mathrm{ht}(p)$), and every model of T_S to a model of C_S (put $\mathrm{sh}_H(p, u, v) := \mathrm{sh}(p, u)$ and $\mathrm{ht}_S(p, v) := v$). So C_H and C_S are conservative over T_H and T_S, whose consequences coincide by (a). The interventional consequences are instances of the added axioms: in C_H, $\mathrm{sh}_H(p_1, 1, 2a_1) \cdot 1 = 2a_1$ and $\mathrm{ht}_S(p_1, v) = \mathrm{ht}(p_1) = a_1$; in C_S, $\mathrm{sh}_H(p_1, 1, 2a_1) = \mathrm{sh}(p_1, 1) = a_1$ and $\mathrm{ht}_S(p_1, v) = v$. If $a_1 \neq 0$ then $2a_1 \neq a_1$, so C_S refutes the datum $\mathrm{sh}_H(p_1, 1, 2a_1) = 2a_1$ that C_H proves. □

Computations. A computation (code in the accompanying repository) confirms the counts by exhaustive search over all subsets of finite instance pools, with exact linear algebra over the rationals, on 500 random data sets with one to four poles (shadows at $u = 2$, and at $u = 3$ in a second implementation); on 2000 random data the four theories prove, refute or leave open exactly the same data. In the literal syntax of that code (numerals counted as one token), $|\Lambda_s| - |\Lambda_h| = 20$ tokens and $\Delta_{\mathrm{sh}} = 1$ token, so T_S is longer than T_H by one token per pole.

A.6 The flagpole with a system of laws

Kitcher's answer to the asymmetry appeals to other laws in which the height figures, which the toy of Proposition A.7 lacks. We add such laws.

Setting. As in Appendix A.5, with further function symbols $\mathrm{qt}_1, \ldots, \mathrm{qt}_k$ from poles to reals (further quantities: weight, the height on the construction plan, and so on), nonzero rational numerals $\theta_1, \ldots, \theta_k$, and data $\mathrm{qt}_j(p_i) = v$. The laws in height form and in shadow form are $\Lambda_j^{\mathrm{ht}} := \forall p\, \mathrm{qt}_j(p) = \theta_j \cdot \mathrm{ht}(p)$ and $\Lambda_j^{\mathrm{sh}} := \forall p\, \mathrm{qt}_j(p) = \theta_j \cdot \mathrm{sh}(p, 1)$. The theories are the height hub $T_H^k := \Lambda_h + \{\Lambda_j^{\mathrm{ht}}\}_{j \le k} + \{\mathrm{ht}(p_i) = a_i\}_i$, the mixed theory $T_S^k := \Lambda_h + \{\Lambda_j^{\mathrm{ht}}\}_j + \{\mathrm{sh}(p_i, 1) = a_i\}_i$, and the shadow hub $T_{SS}^k := \Lambda_s + \{\Lambda_j^{\mathrm{sh}}\}_j + \{\mathrm{sh}(p_i, 1) = a_i\}_i$.

Dark variant. Add a predicate Sunlit on poles, restrict the shadow law to sunlit poles, $\Lambda_h^{\mathrm{D}} := \forall p\, (\mathrm{Sunlit}(p) \to \forall u\, (u > 0 \to \mathrm{sh}(p, u) \cdot u = \mathrm{ht}(p)))$, and take sunlit poles $p_1, \ldots, p_{N_\mathrm{p}}$ and dark poles $p'_1, \ldots, p'_{N_\mathrm{d}}$, with the axioms $\mathrm{Sunlit}(p_i)$ and $\neg\mathrm{Sunlit}(p'_l)$ in every theory and rational numerals a'_l for the heights of the dark poles. Laws in shadow form must split cases:

$$\Lambda_j^{\mathrm{sh,D}} := \forall p \Big(\big(\mathrm{Sunlit}(p) \to \mathrm{qt}_j(p) = \theta_j \cdot \mathrm{sh}(p, 1)\big) \wedge \big(\neg\mathrm{Sunlit}(p) \to \mathrm{qt}_j(p) = \theta_j \cdot \mathrm{ht}(p)\big) \Big).$$

The theories are $T_H^{k,\mathrm{D}} := \Lambda_h^{\mathrm{D}} + \{\Lambda_j^{\mathrm{ht}}\}_j + \{\mathrm{ht}(p_i) = a_i\}_i + \{\mathrm{ht}(p'_l) = a'_l\}_l$ and $T_{SS}^{k,\mathrm{D}} := \Lambda_h^{\mathrm{D}} + \{\Lambda_j^{\mathrm{sh,D}}\}_j + \{\mathrm{sh}(p_i, 1) = a_i\}_i + \{\mathrm{ht}(p'_l) = a'_l\}_l$, each with the Sunlit axioms. A *literal code* charges a theory by the number of symbols it writes, each axiom separately, without definitions; $|\cdot|$ denotes this length, a symbol qt_j counts as one token, and $\Delta_{\mathrm{sh}} := |\mathrm{sh}(x, 1)| - |\mathrm{ht}(x)|$ is the extra length of the reference-shadow term ($\Delta_{\mathrm{sh}} = 0$ if the language has a primitive for the reference shadow).

Proposition A.9 (The flagpole with a system of laws)**.** *(a)* Class tie. *T_H^k, T_S^k and T_{SS}^k are deductively equivalent over B, and so are $T_H^{k,\mathrm{D}}$ and $T_{SS}^{k,\mathrm{D}}$; so for every k no data set separates them, and their deductive classes coincide. (b)* Universal prior. *$|K(T_{SS}^k) - K(T_H^k)| \le c$ and $|K(T_{SS}^{k,\mathrm{D}}) - K(T_H^{k,\mathrm{D}})| \le c$, with c independent of k, N_p, N_d and the numerals. (c)* Literal codes. *$|T_{SS}^k| - |T_H^k| = (|\Lambda_s| - |\Lambda_h|) + (k + N_\mathrm{p})\Delta_{\mathrm{sh}}$. In the dark variant, $|T_{SS}^{k,\mathrm{D}}| - |T_H^{k,\mathrm{D}}| = k\,\Delta_{\mathrm{split}} + \sum_{j \le k} |\theta_j| + N_\mathrm{p}\Delta_{\mathrm{sh}}$, where $\Delta_{\mathrm{split}} > 0$ is the length of the case-split frame and $|\theta_j|$ the length of the numeral θ_j, which the split law writes twice; so the shadow hub loses by at least $k\,\Delta_{\mathrm{split}}$, an amount growing with k, even when $\Delta_{\mathrm{sh}} = 0$. With one definition the overhead of the laws is a constant: define* $\mathrm{z}(p)$ *once as* $\mathrm{sh}(p, 1)$ *on sunlit poles and* $\mathrm{ht}(p)$ *on dark ones, and write every law in* z*; in every model of the theory,* z *is coextensive with* ht*. (d)* Derivation charges *(instances of universally quantified conjuncts, as in Appendix A.5). A datum $\mathrm{qt}_j(p_i) = \theta_j a_i$ has least cost* 1 *for T_H^k and T_{SS}^k, and* 2 *for the mixed theory T_S^k. On shadow data with $u \ne 1$ the two hub theories have least cost* 1 *each; on heights T_H^k has* 0 *and T_{SS}^k has* 1*; on reference shadows the reverse.*

Proof. (a) Each theory contains a form of the shadow law, which gives $B \vdash \mathrm{ht}(p) = \mathrm{sh}(p, 1)$ for every pole p (for every sunlit pole in the dark variant), as in the proof of Proposition A.7(a). Substituting, each law in shadow form is equivalent to the law in height form, and each reference-shadow initial condition to the corresponding height condition. In the dark variant, the case split collapses to $\mathrm{qt}_j(p) = \theta_j \cdot \mathrm{ht}(p)$, because on sunlit poles $\mathrm{sh}(p, 1) = \mathrm{ht}(p)$ by Λ_h^{D} at $u = 1$.

(b) The syntactic translations and their inverses are fixed computable maps on axiom lists: from T_H^k to T_{SS}^k, replace Λ_h by Λ_s, each Λ_j^{ht} by Λ_j^{sh} and each $\mathrm{ht}(p_i) = a_i$ by $\mathrm{sh}(p_i, 1) = a_i$; from $T_H^{k,\mathrm{D}}$ to $T_{SS}^{k,\mathrm{D}}$, replace each Λ_j^{ht} by $\Lambda_j^{\mathrm{sh,D}}$ (reading θ_j and qt_j off the axiom) and the initial condition of each pole declared sunlit by its reference-shadow form. So a program for one list

yields a program for the other at a constant extra length, independent of k, N_{p}, N_{d} and the numerals.

(c) Counting symbols: each Λ_j^{sh} differs from Λ_j^{ht} by one occurrence of the replaced term, and so does each sunlit initial condition; this gives $(k + N_{\mathrm{p}})\Delta_{\mathrm{sh}}$ besides the law texts. Each $\Lambda_j^{\mathrm{sh,D}}$ is longer than Λ_j^{ht} by a fixed frame of length $\Delta_{\mathrm{split}} > 0$ (the connectives, the two Sunlit atoms, a second copy of $\mathrm{qt}_j(p) = \cdot$, and the term $\mathrm{sh}(p, 1)$) and by a second copy of the numeral θ_j; both dark theories contain Λ_h^{D}, the dark initial conditions and the Sunlit axioms. The definition of z is a single axiom of constant length, and laws written in z have the length of the laws written in ht. In every model of the theory, $\mathrm{z}(p) = \mathrm{sh}(p, 1) = \mathrm{ht}(p)$ on sunlit poles by Λ_h^{D} at $u = 1$, and $\mathrm{z}(p) = \mathrm{ht}(p)$ on dark poles by the definition.

(d) For each pole the instance graphs are trees: for T_H^k and T_S^k, a star centred at $\mathrm{ht}(p_i)$ with leaves $\mathrm{sh}(p_i, u)$ and $\mathrm{qt}_j(p_i)$; for T_{SS}^k, a star centred at $\mathrm{sh}(p_i, 1)$ with leaves $\mathrm{sh}(p_i, u)$ $(u \neq 1)$, $\mathrm{ht}(p_i)$ and $\mathrm{qt}_j(p_i)$. Lemma A.8 gives the counts: $\mathrm{qt}_j(p_i)$ is at distance 1 from the fixed centre in T_H^k and T_{SS}^k, and at distance 2 from the fixed leaf $\mathrm{sh}(p_i, 1)$ in T_S^k; the shadow, height and reference-shadow counts are those of Proposition A.7(b) for T_H and T_S'. □

References

Wolfgang Balzer, C. Ulises Moulines, and Joseph D. Sneed. *An Architectonic for Science: The Structuralist Program*, volume 186 of *Synthese Library*. D. Reidel, Dordrecht, 1987.

Nuel D. Belnap, Jr. and Thomas B. Steel, Jr. *The Logic of Questions and Answers*. Yale University Press, New Haven, 1976.

José M. Bernardo. Expected information as expected utility. *Annals of Statistics*, 7(3):686–690, 1979. doi: 10.1214/aos/1176344689.

Sylvain Bromberger. Why-questions. In Robert G. Colodny, editor, *Mind and Cosmos: Essays in Contemporary Science and Philosophy*, pages 86–111. University of Pittsburgh Press, Pittsburgh, 1966.

J. Dorling. Bayesian personalism, the methodology of scientific research programmes, and Duhem's problem. *Studies in History and Philosophy of Science*, 10(3):177–187, 1979.

Michael Friedman. Explanation and scientific understanding. *Journal of Philosophy*, 71(1):5–19, 1974.

Clark Glymour. Two flagpoles are more paradoxical than one. *Philosophy of Science*, 45(1): 118–119, 1978.

Jeroen Groenendijk and Martin Stokhof. *Studies on the Semantics of Questions and the Pragmatics of Answers*. PhD thesis, Universiteit van Amsterdam, 1984.

Hans Halvorson. What scientific theories could not be. *Philosophy of Science*, 79(2):183–206, 2012.

Norwood Russell Hanson. *Patterns of Discovery: An Inquiry into the Conceptual Foundations of Science*. Cambridge University Press, Cambridge, 1958.

Carl G. Hempel and Paul Oppenheim. Studies in the logic of explanation. *Philosophy of Science*, 15(2):135–175, 1948.

Jaakko Hintikka. *Inquiry as Inquiry: A Logic of Scientific Discovery*, volume 5 of *Jaakko Hintikka Selected Papers*. Kluwer, Dordrecht, 1999.

Philip Kitcher. Explanatory unification. *Philosophy of Science*, 48(4):507–531, 1981.

Philip Kitcher. Explanatory unification and the causal structure of the world. In Philip Kitcher and Wesley C. Salmon, editors, *Scientific Explanation*, volume 13 of *Minnesota Studies in the Philosophy of Science*, pages 410–505. University of Minnesota Press, Minneapolis, 1989.

Thomas S. Kuhn. *The Structure of Scientific Revolutions*. University of Chicago Press, Chicago, 2 edition, 1970.

Thomas S. Kuhn. Second thoughts on paradigms. In *The Essential Tension: Selected Studies in Scientific Tradition and Change*, pages 293–319. University of Chicago Press, Chicago, 1977.

Larry Laudan. *Progress and Its Problems: Towards a Theory of Scientific Growth*. University of California Press, Berkeley, 1977.

Hannes Leitgeb and Richard Pettigrew. An objective justification of Bayesianism I: Measuring inaccuracy. *Philosophy of Science*, 77(2):201–235, 2010.

Sebastian Lutz. What was the syntax–semantics debate in the philosophy of science about? *Philosophy and Phenomenological Research*, 95(2):319–352, 2017.

Richard Pettigrew. *Accuracy and the Laws of Credence*. Oxford University Press, Oxford, 2016.

Michael Polanyi. *Personal Knowledge: Towards a Post-Critical Philosophy*. University of Chicago Press, Chicago, 1958.

Wesley C. Salmon. *Statistical Explanation and Statistical Relevance*. University of Pittsburgh Press, Pittsburgh, 1971. With contributions by Richard C. Jeffrey and James G. Greeno.

Wesley C. Salmon. *Scientific Explanation and the Causal Structure of the World*. Princeton University Press, Princeton, NJ, 1984.

Joseph D. Sneed. *The Logical Structure of Mathematical Physics*, volume 35 of *Synthese Library*. Reidel, Dordrecht, 1971.

Ray J. Solomonoff. A formal theory of inductive inference, parts I and II. *Information and Control*, 7:1–22 and 224–254, 1964.

Michael Strevens. The Bayesian treatment of auxiliary hypotheses. *British Journal for the Philosophy of Science*, 52:515–537, 2001.

Frederick Suppe, editor. *The Structure of Scientific Theories*. University of Illinois Press, Urbana, 2 edition, 1977.

Patrick Suppes. A comparison of the meaning and uses of models in mathematics and the empirical sciences. *Synthese*, 12(2/3):287–301, 1960.

Patrick Suppes. Models of data. In Ernest Nagel, Patrick Suppes, and Alfred Tarski, editors, *Logic, Methodology and Philosophy of Science: Proceedings of the 1960 International Congress*, pages 252–261. Stanford University Press, Stanford, CA, 1962.

Bas C. van Fraassen. *The Scientific Image*. Clarendon Press, Oxford, 1980.

James Woodward. *Making Things Happen: A Theory of Causal Explanation*. Oxford University Press, New York, 2003.